



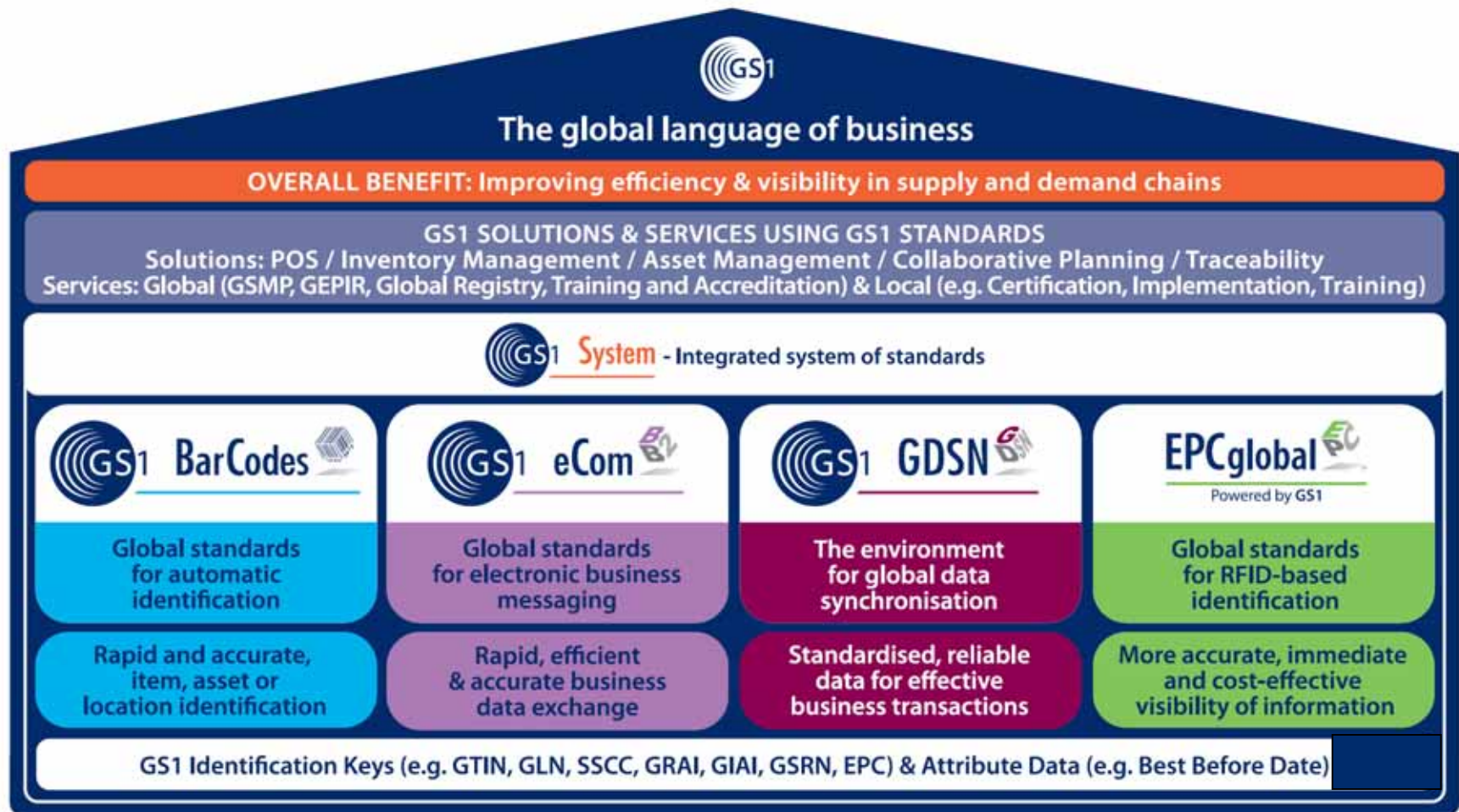
Transport and Logistics Services Business Action Group

Bruno Julien
EPCglobal Inc

Seoul - 23rd of January 2005



GS1's Brand Architecture

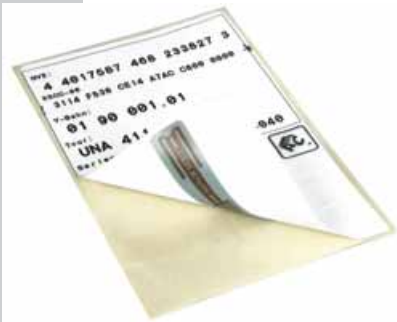




GS1 EPC Global



Global standards for RFID-based identification
More accurate, immediate and cost-effective visibility of information



- A new global standard system that combines radio frequency identification (RFID) technology, existing communications network infrastructure and the Electronic Product Code
- Identify and locate any item in the supply chain of any company, in any industry, anywhere in the world
- Immediate and automatic identification and tracking of an item through the whole supply chain
- Improved efficiency and visibility in the supply chain



EPCglobal Overview

- EPCglobal community developed standards*, guidelines and implementation tools are our deliverables: Enabling mass adoption of the EPC Network across supply chains our goal.
- 2 years old but within long established global standards organization (GS1).
- Based upon Radio Frequency Identification (RFID) which is only one layer in the EPC Network.
- Frequency neutral.
- Multiple industry and expanding.

* Where standards do not exist or fall short of subscriber user requirements.

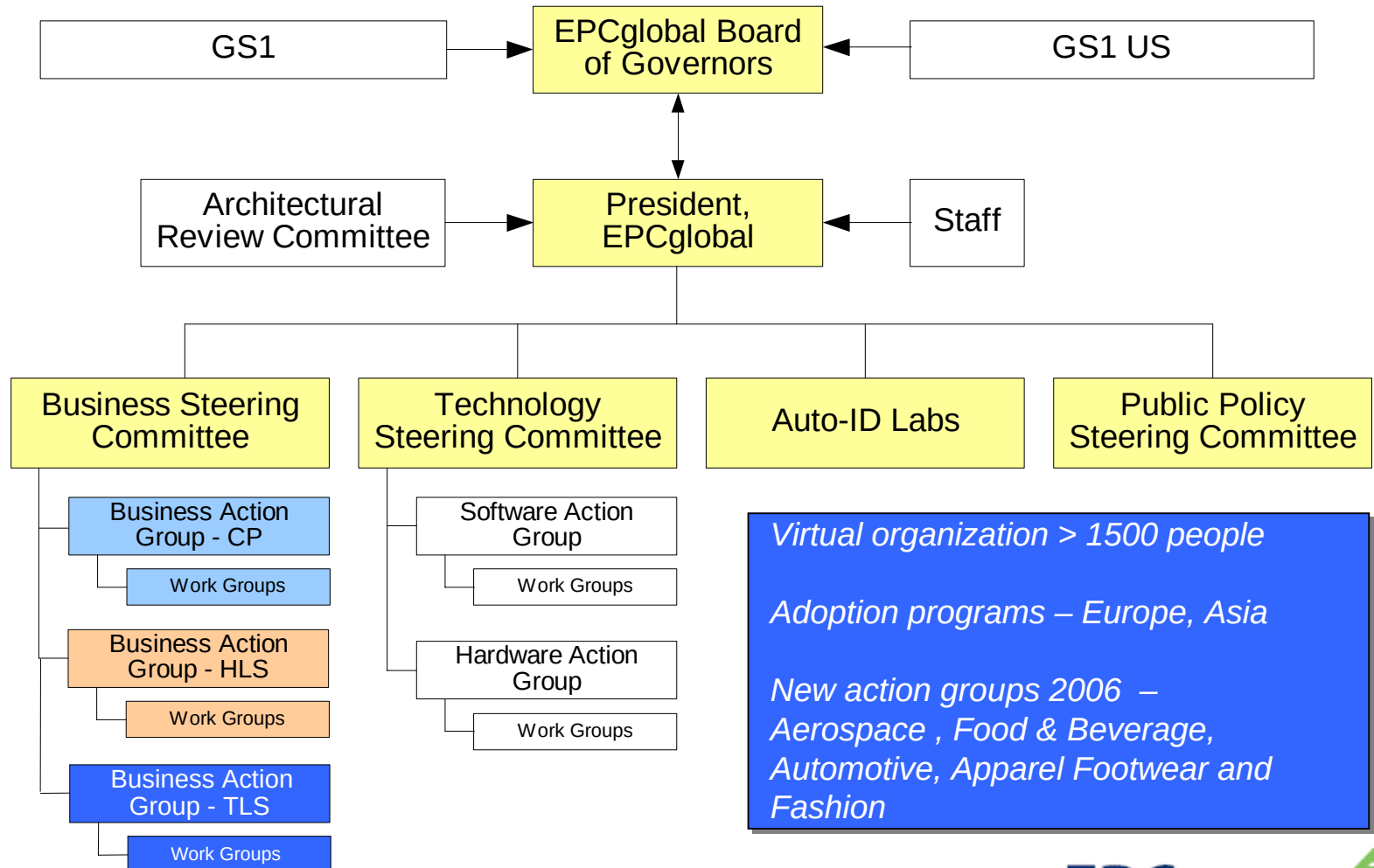


EPCglobal Purpose

- *Global* leadership in developing and promoting *multi-industry, user driven standards* for *collaborative commerce* utilising the *EPC*
- Deliver added value to our customers and stakeholders
- Drive global, multi-industry adoption of EPC via the GS1 Member Organizations
- Act as trusted authority on technical standards relating to use of EPC
- Effectively manage public policy issues relevant to the use of the EPC

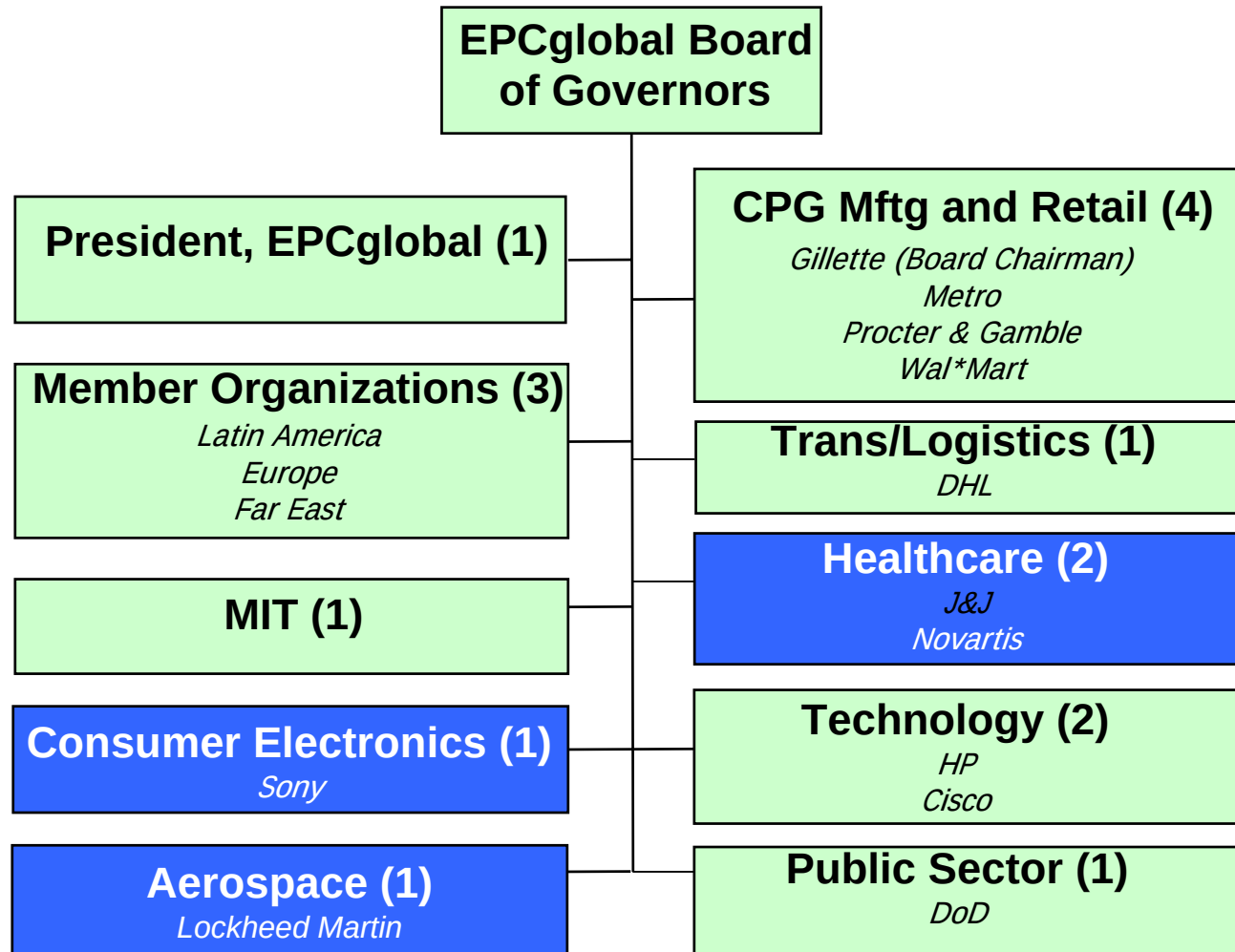


EPCglobal Organization





EPCglobal Board of Governors





EPCglobal research partners – Auto ID Labs

- **Cambridge - UK**
- **MIT – USA**
- **Adelaide - Australia**
- **St. Gallen – Switzerland**
- **Fudan – China**
- **Keio – Japan**
- **ICU – South Korea**





EPC: Membership across the world

	Total # Members	% Total
Asia	148	20%
North America	451	62%
Europe	110	15%
ME & Africa	8	1%
Latin America	16	2%
Total	733	100%

- Each organization joins once in the location of the head office



EPC: Significant growth in all regions

- EPC global is only two years old!

	Jun 2004	Dec 2005	% Increase
Asia	21	148	705%
North America	132	451	342%
Europe	36	110	306%
ME & Africa	2	8	400%
Latin America	0	16	-
Total	191	733	384%

- Note: End 2004 = 391



Most Organizations are at a Similar Point

Determining pathways for competitive advantage

- Focused on identifying priority areas of enablement
- Developing business cases to support budgeting and savings projections
- Implementation pilots to build experience and capabilities
- Contemplating/scheduling customer/supplier initiatives
- Optimization activities for greater clarity in calculating return on investment
- Increase in commercial viability as standards are developed and lead to continued decreases in costs



Role of EPCglobal

Supporting industry transformation

- Significant industry initiatives require neutral leadership to shepherd transformation process
 - EDI, barcodes, product identification, data synchronization, etc
- Industry priorities and needs must be addressed to meet timelines and avoid excessive costs
 - “By industry, for industry” is the crucial success factor
 - Avoidance of fragmented approaches is a must
 - All efforts must support “supply chain convergence”
- Only successful path forward is a global approach
 - Leverage reach of committed companies and supply chains
 - Directly engage other industries, associations, standards bodies, and governments (i.e. ISO, China, etc.)
- Efforts must consider the coexistence, integration and transition from/with today’s successful bar code implementations



Background to TLS BAG

- **Original initiative by DHL and Exel**
- **TLS keen to develop total industry solution rather than deliver one solely for FMCG / Retail**
- **Ensure consideration of TLS needs in global standards development**
- **Ensure interoperability to verticals**
- **Ensure interoperability within TLS**
- **Promote TLS adoption**



Relevance to TLS Industry

Key challenges facing industry:

- **Supply Chain Efficiency**
 - Inventory Visibility / Out of Stocks
 - Error Reduction
 - Theft / Diversion
 - Factor utilization and productivity
- **Asset Visibility / Prevention of Loss / Theft**
 - Logistics Units (containers, pallets, ...)
 - Tools and Equipment
 - Property / Yard Management
- **“Safe and secure” Initiatives**
 - Anti-Counterfeit
 - Health & Safety
 - Homeland Security (US)
 - Food Supply

RFID as enabler of:

- **Competitive advantage**
- **Asset Management**
- **Compliance**



History TLS Formation

Brussels Summit

- **March 05**
- **Hosted by DHL**
- **Approx 50 attendees from 23 Companies**
 - Representation from Europe, Asia & US
 - Representation of all sectors, Shipping, Airlines, Freight Forwarders, 3PL's, Express Parcel, Transport
- **Industry priorities discussed**
- **Regular teleconferences established**

Singapore Summit

- **July 05**
- **Hosted by NYK**
- **Approx 60 attendees from c25 companies**
 - Greater Asian presence
- **TLS Charter agreed**
- **Work groups defined:**
 - **Transportation**
 - **4 Walls**
 - **Import / Export**
 - **Integration**
- **Co-chairs appointed**

Atlanta Summit

- **September 05**
- **Pre-meeting to EPCglobal US conference**
- **Approx 50 attendees from c25 companies**
 - Less American representation than expected
- **Work group charters agreed**
- **Work plans drafted**
- **Formal approval initiated**



Objective TLS

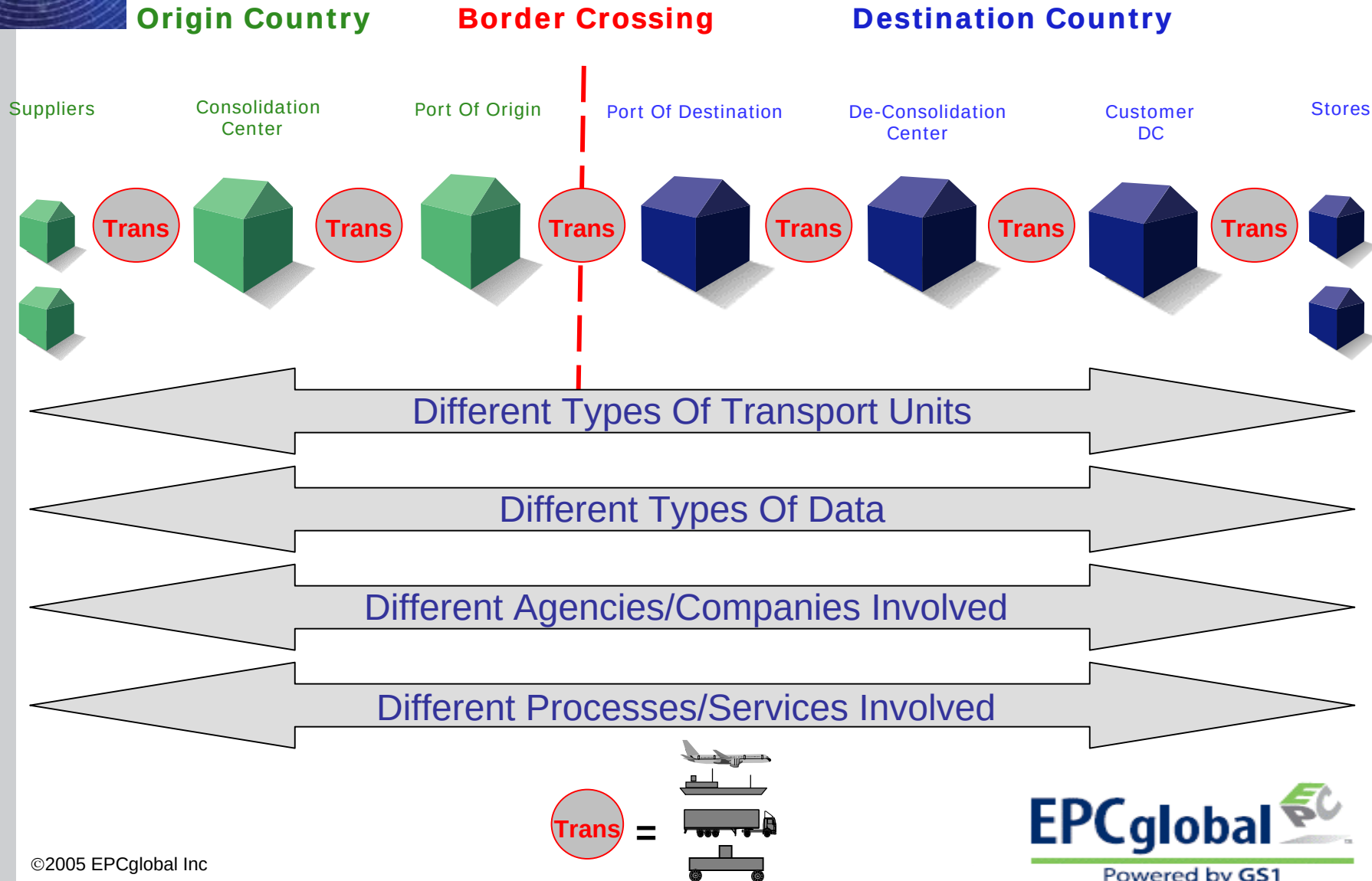
Develop or review global standards specifications for transportation and logistics services

Standards must:

- **Comply with, fulfill and add measurable value to the business objectives defined by:**
 - User groups, trading partners, and customers
 - Other Business Action Groups
- **Improve the efficiency of:**
 - Transport, logistics and services providers
 - Supply chains. Solution development should consider the most cost effective solution for the specific supply chain or product application.

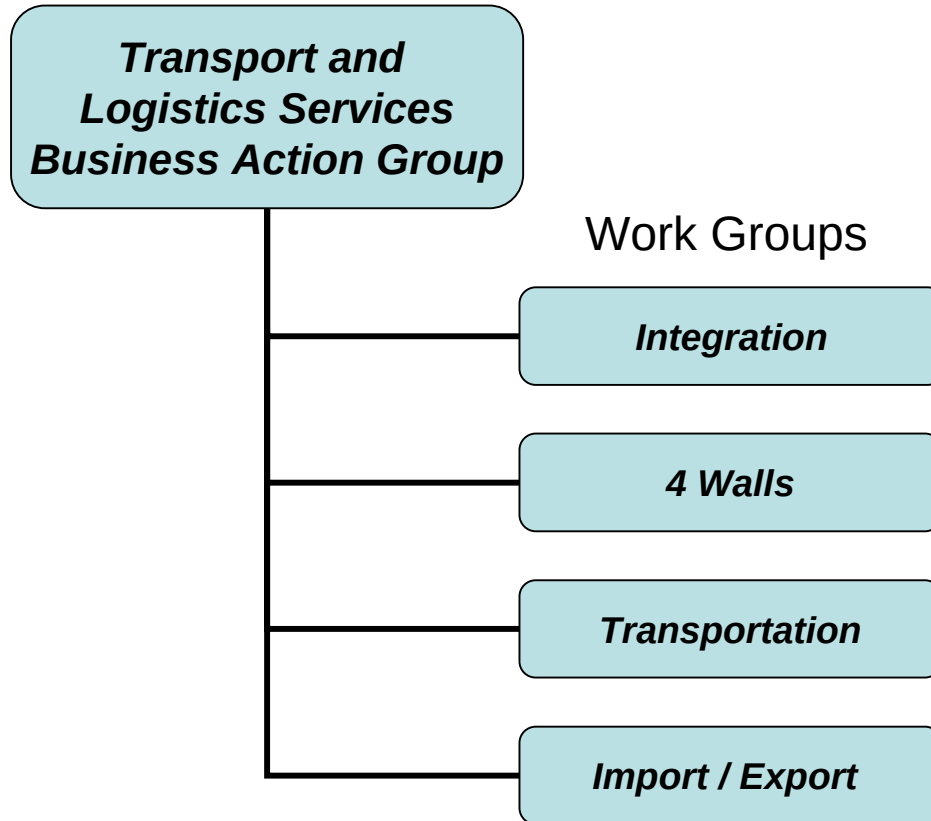


The Global Chain of Custody





TLS Work Groups



Output

- Define use cases
- Define unique requirements
- Map Use Cases and Requirements
- Present harmonized Use Cases and Requirements to Industry BAG, Hardware Action Group (HAG) and Software Action Group (SAG)



TLS Work Groups

- **Transportation**

Identify benefits of EPC RFID in transportation and develop specifications (Standards) to enable organizations to improve visibility, performance, security, quality and communications

- **4 Walls**

Deliver business functional requirements / specifications related to processes that occur within the 4 walls.

- **Integration**

Identify and present a collaboratively developed set of generic standards governing the format continuity and types of data traversing the global supply chain permitting visibility to the chain of custody at each transport unit level.

- **Import / Export (Customs)**

Streamline import and export procedure by mean of RFID.



First official BAG meeting

- From January 15th to 18th 2006 in Kobe (Japan)
- 110 participants from 60 companies
- Hosted by NYK with a pilot demonstration
- Every work group had separate and joined sessions
- Creation of a Strategy group

Next meeting : 17th and 18th of Mai in California

You are welcome



Support from other organizations

- International Air Transport Association (IATA)
 - Key note presentation at TLS BAG in Kobe
 - Commitment to build an alliance. Further discussion planned.
- World Customs Organization (WCO)
 - Developing upon positive response to initial contact
 - Several Customs authorities have pledged support when WCO recognition received
- Chartered Institute of Logistics and Transport (CILT)
 - Representative selected and to join next TLS BAG



IATA key note presentation, Kobe



Working Together

- Does it make sense for cargo shipments (packets) to have an EPC tag from the manufacturer of the goods and from the shipper... and from the airline?
- EPCglobal will have a rich network of information.
- Many tagged items will be destined for loading on aircraft.
- There should be some way of sharing this information to facilitate airport processes – e.g.
 - Tagging of dangerous goods
 - Shipping of packets.
 - Loading of hold cargo.
- If we know what and where it is, we can tell you.



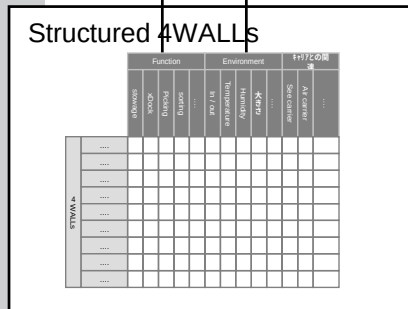
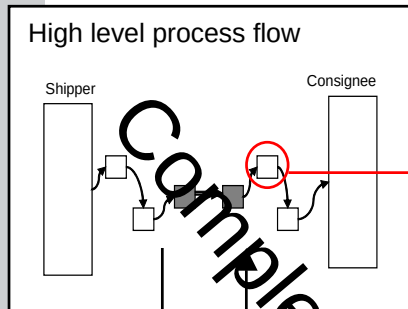
Accomplishments While In Kobe

- An integration framework has been roughly established
- Critical data elements are starting to be identified
- Work-groups are considering supply chain benefits
- Focus on developing a 2006 Work-plan
- Identification of issues/opportunities
- Discussions about future solutions / pilots
- Identification of the need to engage other parties such as government agencies and other logistics bodies (WCO, WSC)
- Continued alignment of work-groups and identification of key leadership candidates

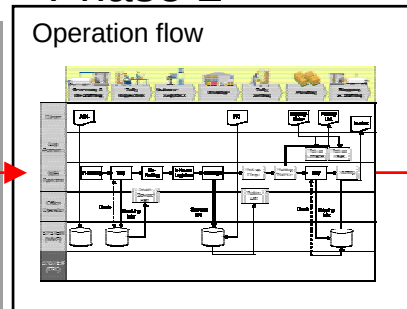


4 Walls Working Group Phases

Phase 1

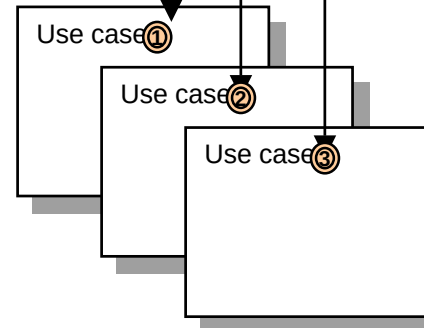
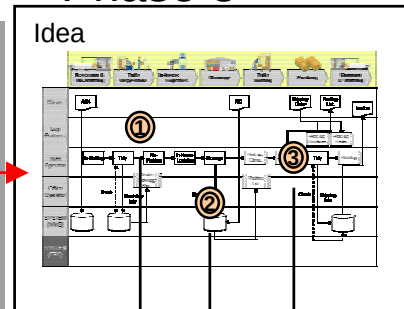


Phase 2



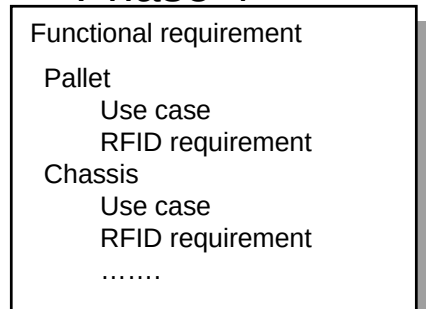
Identify
Critical
“Hand Offs”

Phase 3



Create
Detail flows for
“Hand Offs”

Phase 4



Identify
TAG Functional
Requirements I.e.
Distance, Time
Environment, etc.



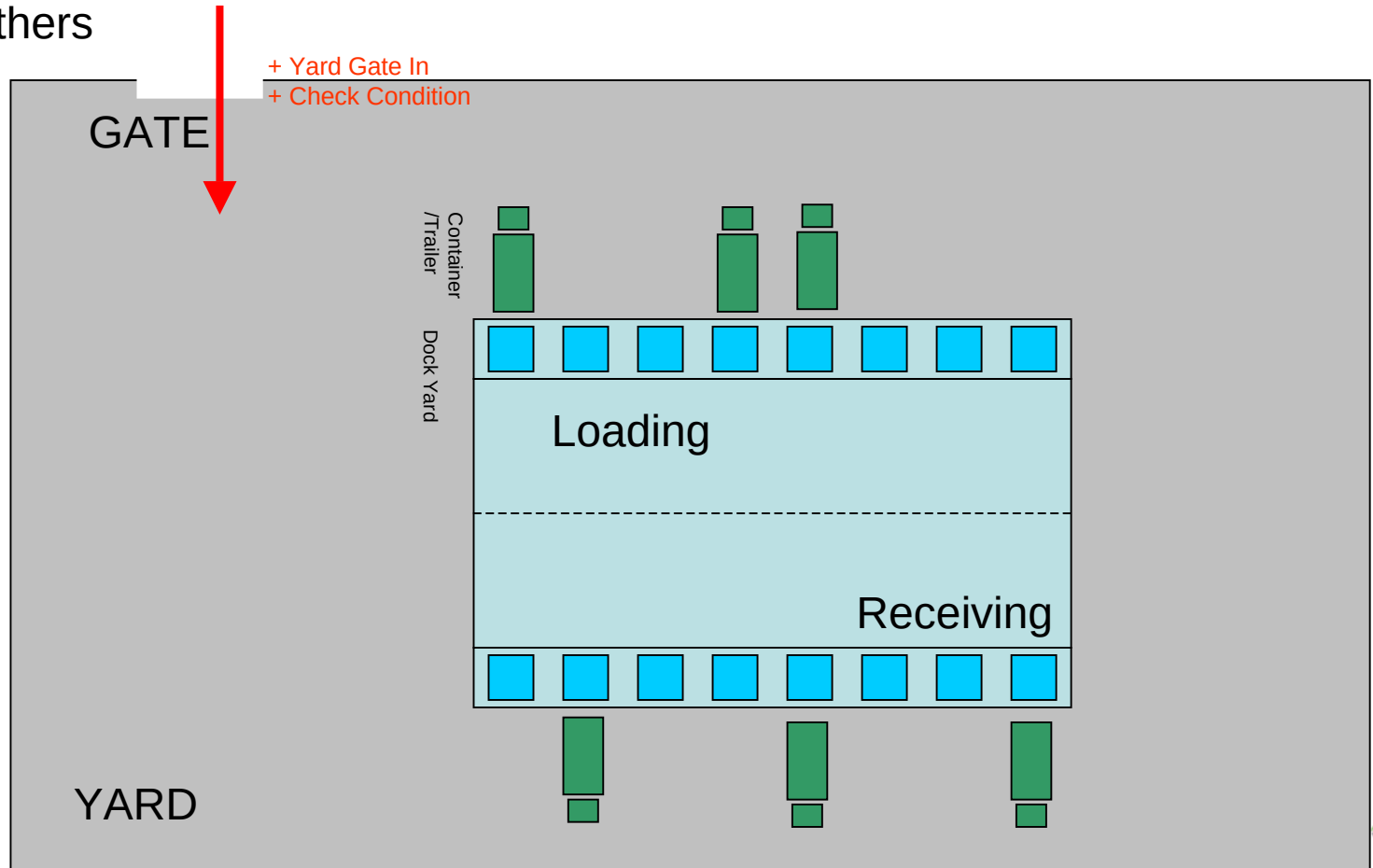
4 Walls Working Group Work Plan / Accomplishment

- Phase 1 – Develop the high level “standard/generic flow” for a 4 Walls facility (Complete)
- Phase 2 – Determine all the “detailed handoffs”, this is a more detailed flow of each of the critical handoffs from the high level flow.
 - 4 Teams Established- Inbound, Outbound, Storage and Yard Management
 - Leaders being Identified
- Phase 3 – At all levels of flow and handoff’s determine what information / Data is needed
- Phase 4 – given phase 1-3 above, now determine what type of RFID tag is recommended, considering the working environment, time and space



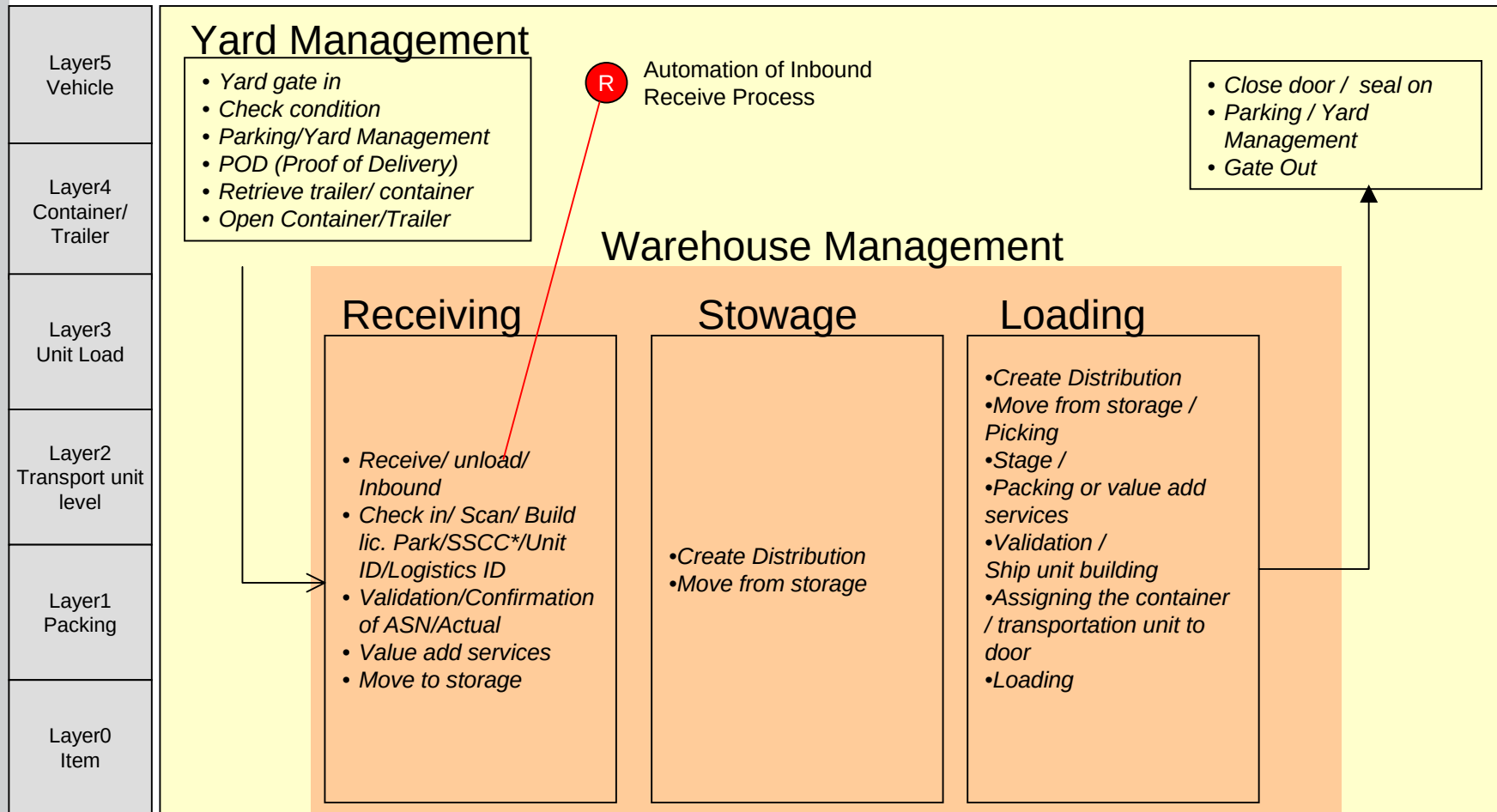
4 walls Areas of Responsibility

- Overlap w/ Transportation in Yard Mngt. Area, work together on this
- Unresolved Area is ownership of PU & Delivery (POD), work w/ others





Target of Possible Pilot, Based on Issues / Opportunities Identified from Team





Import / Export : Benefit for customs clearance procedure

- Security
- Accuracy
 - Eliminate wrong shipment
 - Reliability of data
 - Better identification
 - Reduce human error
- Real-time monitoring (Tracking and Tracing)
- Improve timeline
- Cost saving



Import / Export : Work plan for 2006

Objective;

- 1 Develop Practical Business Model
- 2 Plan and Implement Simple Pilot Case to Show Benefit

Quarterly Action Plan;

1stQ Analyze Data Element in Customs Clearance
Develop Detailed Business Model
Survey Preceded Pilot Case

2ndQ Contact Customs and WCO
Develop Detailed Pilot Plan
Recruit Participant for Simple Pilot Case

3rdQ Preparation for Simple Pilot Case

4thQ Implementation and Evaluation



Transportation Recap

- o Finished supply chain use cases
- o Identified conveyance types (container, trailer, ULD, air pallet, cage, truck)
- o 2006 Goal: A functional specification document describing the business requirements for a tag associated to a conveyance type
- o Developed 'Process – Benefits – Roadblock' review
- o First review with TLS working groups
- o Next steps:
 - o Data requirements within the use cases
 - o Data association with the nested data representing the cargo

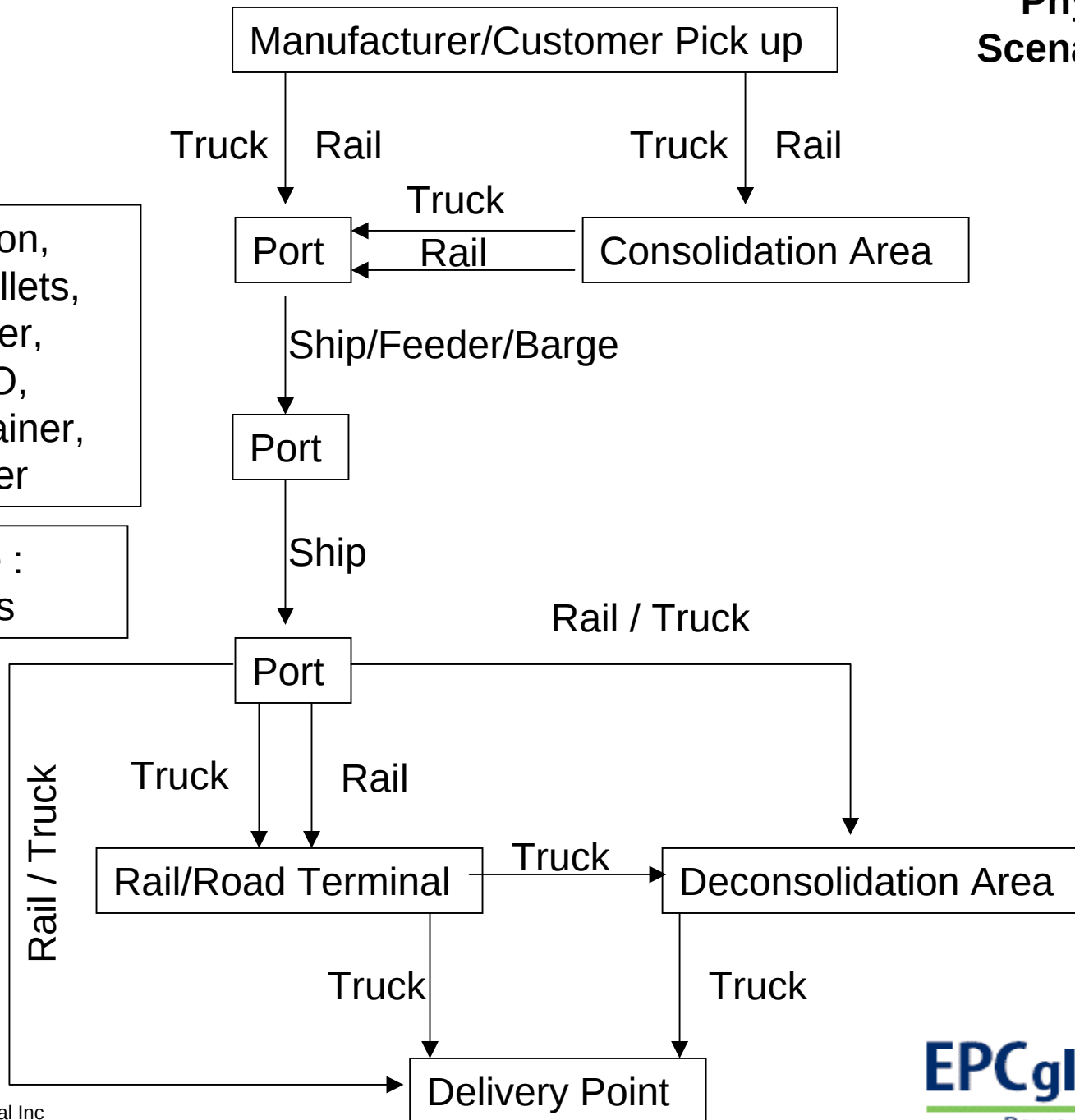
Physical Flow Scenario 1 : Sea



Scope : Carton,
Package, Pallets,
RTI, Container,
Items, Ro-RO,
Ocean Container,
Assets, Trailer

Not in Scope :
Bulk, Chassis

Conveyance
/Transportation:
Container,
Trailer

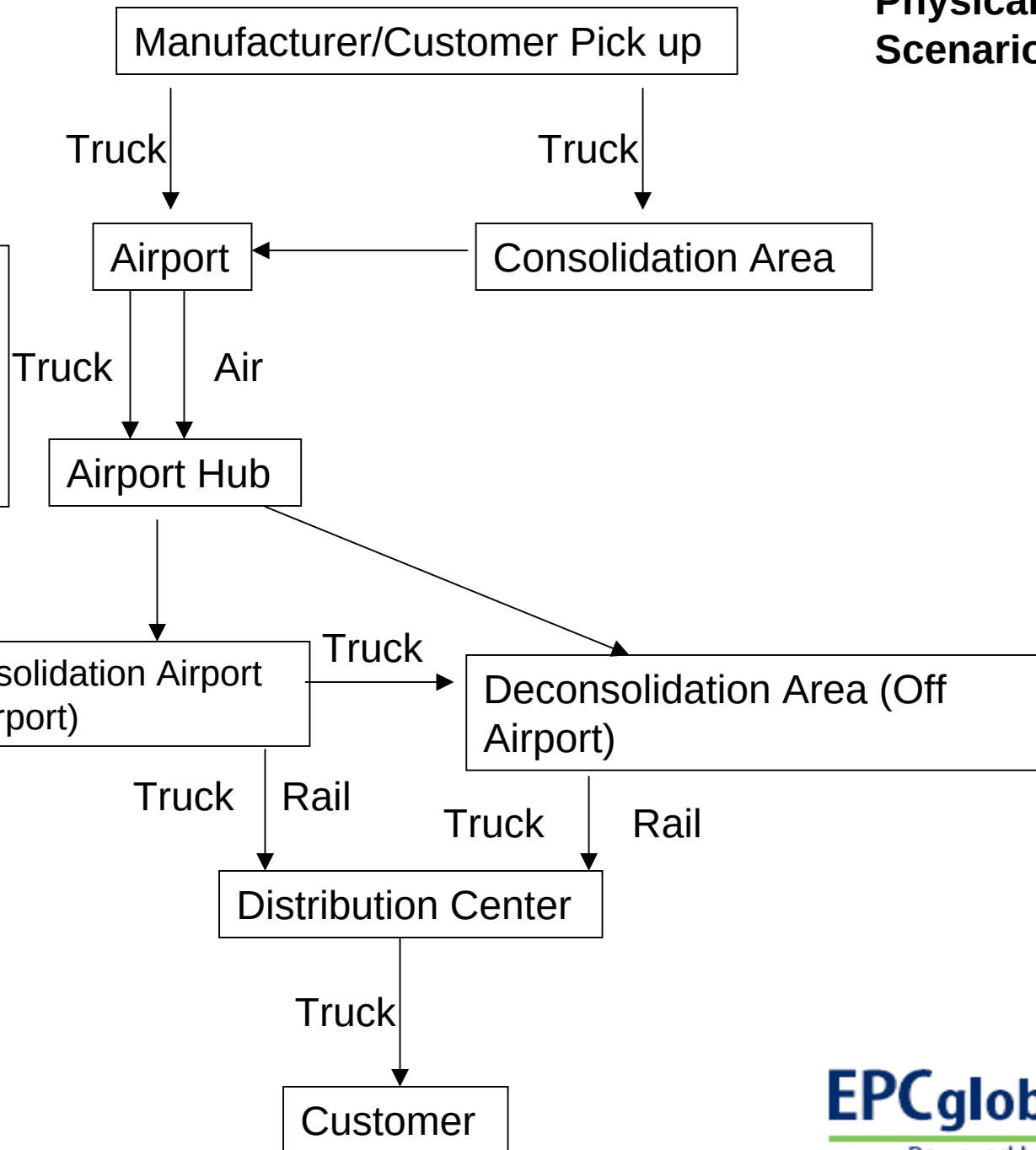


Physical Flow Scenario 2 : Air



Scope : Carton,
Package, Pallets,
Air Pallets, ULDs,
Empty ULDs,
Trailer

Conveyance Air
/Transportation:
ULD, Cage
(Temporary
Transportation
Unit), Air Pallet

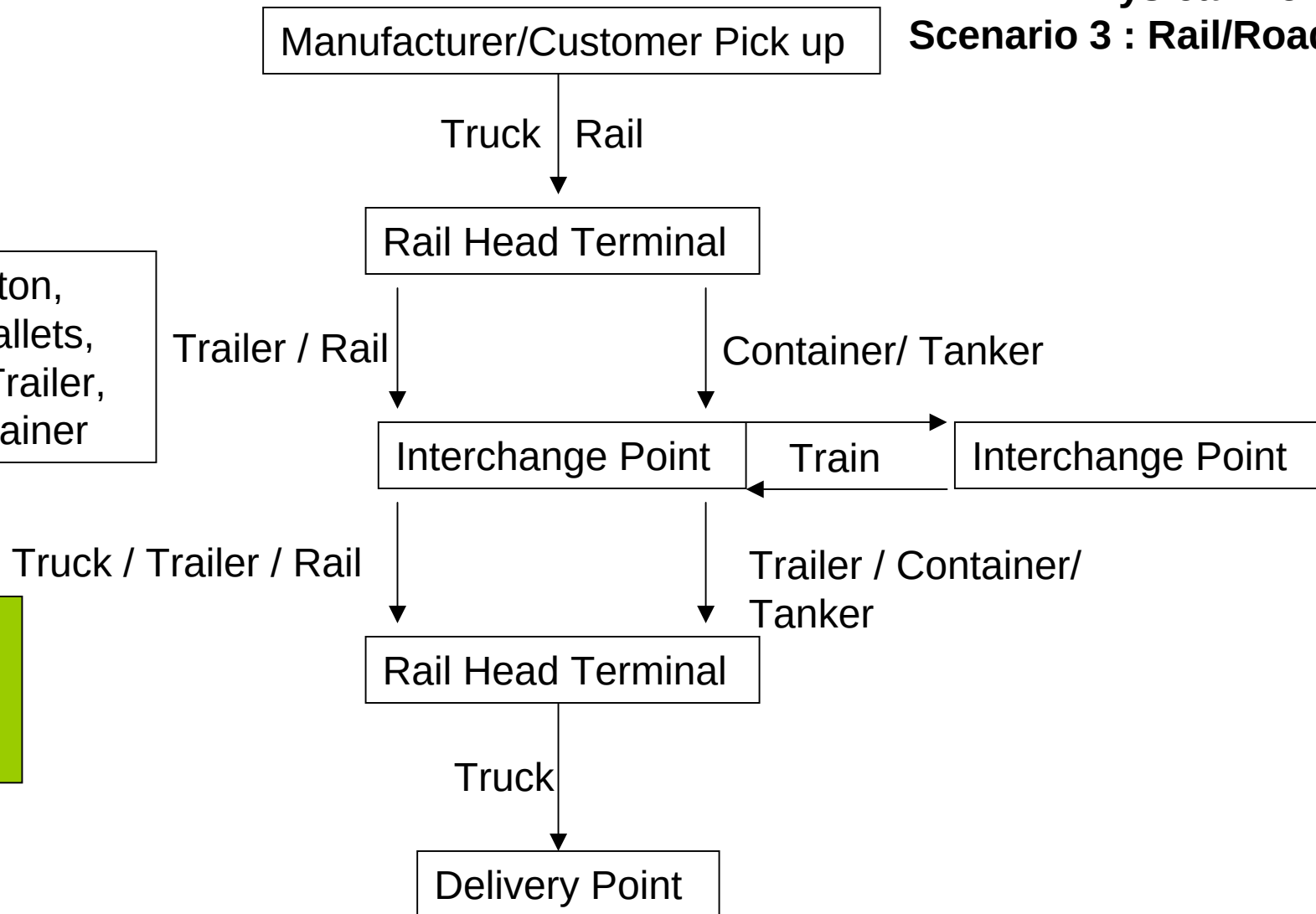




Physical Flow Scenario 3 : Rail/Road

Scope : Carton,
Package, Pallets,
Container, Trailer,
Ocean Container

Conveyance
/Transportation:
Container,
Trailer, Truck



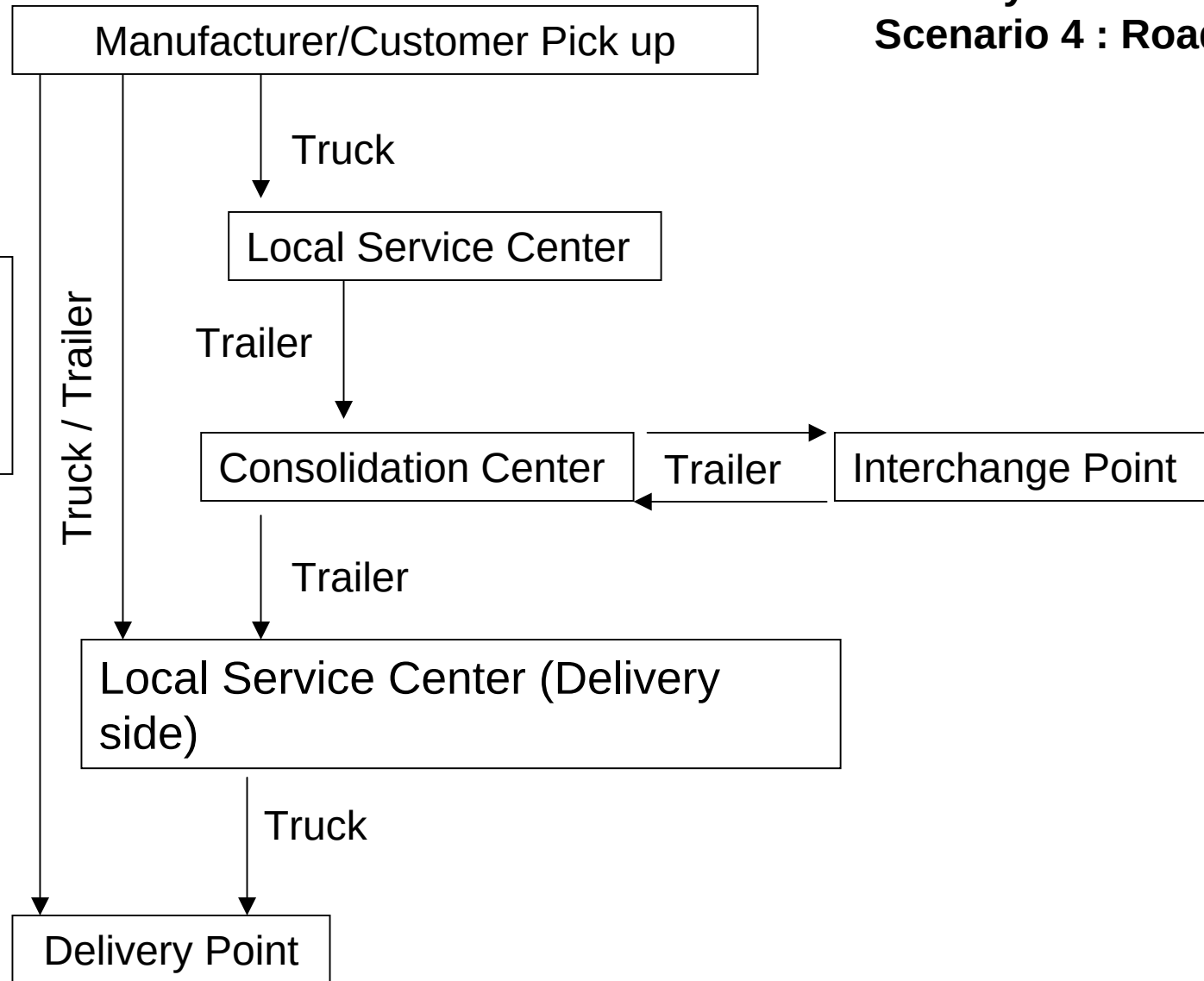


Physical Flow Scenario 4 : Road

Scope : Carton,
Package, Pallets,
Container, Trailer,
Ocean Container

Conveyance
/Transportation:

Trailer, Truck,
Container





Transportation main issue

Adoption of the Technology

- **Identify the key drivers affecting RFID adoption in your area**
- **Identify specific areas and assess likely speed of adoption (early <2, mid-term <5, longer 5+)**



Transportation : Benefits

- o Visibility
- o Change of Custody
- o Velocity (thru-put through a transportation node)
- o Real-time data capture
- o Data error reduction
- o Efficiency
- o Anti-theft
- o Cargo condition
- o Claims reduction



Transportation Roadblocks

- o Cost
- o Global industry adoption
- o Regulatory e.g. frequency differences
- o Global variable cost of labor
- o Mandates
- o Standards



Integration Scope

- Terminology Definition
- Project Team Development
- Tool Selection
- Scope Definition
- Business Requirements Definition
- Architecture
- Pilot Implementation



Integration Priorities

- Define visibility characteristics
- Define chain of custody
- Initiate Overall Project/Program Management Approach
- Describe Chain Of Custody Scenario
- Formalize Agreement On Chain Of Custody
- Establish Pilot Goals
- Creation Of Template (Includes Definition Of Guidelines)
- Capture Inputs Of Other TLS Workgroups
- Define Specific Business Processes
- Validate critical events for visibility
- Develop Use Cases
- Validate data Capture To Support Events and common data elements



We still need more TLS members

Key players:

- **Carriers: Ocean, Air, Rail, Road**
- **Integrators: Express, Mail**
- **Freight Forwarders (multi-modal)**
- **Logistics Service providers (3PL / 4PL)**
- **Port / terminal operators**
- **Multiples**

Characteristics:

Asset based
Network / Asset based
Non-asset based
Contract based
Asset based
-



Transport & Logistics Services

•Why should people attend

- To be involved in identifying solutions that should work for their business
- To drive open standards to enable market competition for product
- To leverage the technology members who will work with their requirements rather than selling a product to them
- To understand how other people plan to successfully deploy the technology
- To understand what role EPCglobal play in developing standards and how the EPCglobal Network will add value in their supply chains
- To ensure that the total needs of the industry and the sectors it serves are represented.

•How are people already using the technology

- Asset Utilisation
- Increased Operational Efficiency
- Health/Safety & Security
- Quality Control and Customer Care
- Financial Management
- Service Offering to Comply with Mandates
- Repair & Maintenance Scheduling
- Consignment Tracking



In summary...

- *EPCglobal community.....from hope to hype to implementation.*
 - The EPCglobal community grows stronger every day
 - Life is becoming more complex.... but this is a good sign!
 - Physics and standards challenges are being overcome
 - Prices are going down
 - The benefits of “visibility” are starting to be realised
 - The community has moved decisively from preparation to implementation

The race for business benefits and competitive advantage is on!!



Questions?



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